Ticking Bomb: Game Design Document

Oct 14, 2026 · @David Olamide

1. Overview

A ball-shaped bomb ticks down while players race to solve physical puzzles on it, and whoever's holding it when time runs out gets blown up. It's a loud, social hardware party game built for events and festivals like Amaze and GDC.

The pitch in one breath: The screen calls a color. The player wearing that color grabs the bomb. Solve the puzzle on the ball to reset the timer, or gamble on a pass and hope you don't hit the wrong button. If it explodes in your hands, you're out.

Core fun we're protecting:

- Time pressure you can hear and feel
- A ball you desperately want out of your hands
- People shouting at each other to help (or sabotage)
- A spectacle that's fun to watch, not just to play

Event requirements:

- Teachable in under 30 seconds
- Short rounds so the line keeps moving
- Self-contained and portable: the ball is the whole game, no laptop
- Works in loud rooms with bad Wi-Fi (it doesn't use Wi-Fi)
- Quick for one booth host to set up, calibrate, reset and run all day

2. Design pillars and scope

Version 1 ships two modes (Solo and Teams) with five puzzles, one ball and up to four headbands. Imposter mode is designed later.

Pillars

1. **Panic, not paralysis.** Every decision is fast. No menus mid-round.
2. **The ball is the star.** All input happens on the ball. The screen supports, it doesn't replace.
3. **Loud and readable.** Every event is shown on at least two feedback channels.
4. **Tweakable.** Timer lengths, puzzle counts and pass odds are settings, not code changes.

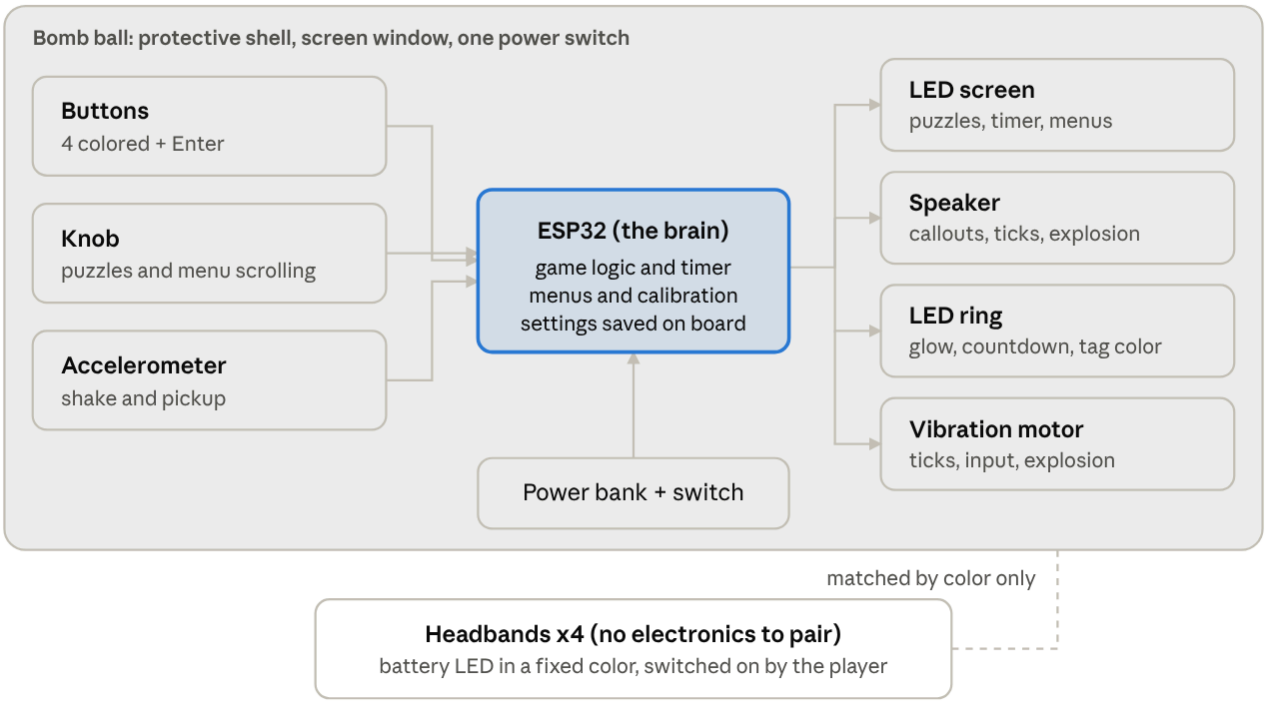
Scope for the one-week build

Area	In v1	Later
Modes	Solo (Rapid Fire), Teams, Individual	Imposter
Puzzles	Maths, number sequence, color sequence, knob to position, shake	Shout at it
Players	Solo: any number around one ball. Teams: 4 headbands	Up to 8 headbands
Electronics	1 ESP32, inside the ball	Smart headbands with their own board
Ball input	4 colored buttons, Enter, knob, accelerometer	Microphone
Feedback	LED screen, speaker, LED ring, vibration, all on the ball	
Headbands	Battery LED in a fixed color, no electronics to pair	Headbands that light up when tagged
Power	One power bank and one switch	Custom LiPo and charging
Handling	Passed hand to hand, screen protected against drops	Throwable, impact-proof
Setup	On-ball menu with knob and Enter	Per-player color selection

3. System architecture

The ball is the whole game. One ESP32 inside it runs the logic and drives the LED screen, speaker, LED ring and vibration motor. No laptop, no base station, no Wi-Fi.

One ESP32 inside the ball runs the whole game



system architecture · laptop, base station, ball, headbands

One board means one firmware to flash, nothing to pair and one switch to start. The whole kit is the ball plus a bag of headbands.

Parts in the ball

Part	Job	What we have	Status
ESP32 board	Runs everything	2 ESP32 dev boards. One is the brain, one is a backup	Have
Puzzle screen	Puzzles, menus, callouts	16x2 LCD with a separate I2C backpack	Have
Timer display	Big, bright countdown	2 quad alphanumeric displays (4 characters each)	Have
Speaker	Beeps, ticking, explosion	1 round speaker	Have

Part	Job	What we have	Status
MP3 module + microSD card	Voice callouts and recorded sounds	Not in the photos. Beeps, ticking and a few short clips can play from the ESP32's memory through the amp. Needed only for many voice and mockery lines	Need to buy
Amplifier module	Drives the speaker	PAM8403 stereo amp board, up to about 3W per channel on 5V, already wired to the speaker. Fed from an ESP32 DAC pin	Have
Addressable LED strip	Glow, countdown, tag color	Several pieces of RGB LED strip, used instead of a ring	Have
Motor driver (transistor or MOSFET + diode)	Lets the ESP32 power the motor safely	Not in the photos	Need to buy
Vibration motor	Ticks, input feedback, explosion	About 4 round motors with offset weights. One is enough for v1	Have
Knob (rotary encoder)	Knob puzzle, menu scrolling	2 rotary encoders with push buttons. One goes in the ball, one is a spare	Have
Accelerometer	Shake puzzle, pickup detection	Not in the photos	Need to buy
4 colored buttons + Enter	Puzzle answers, menu select	About 12 large capped buttons (black, yellow, blue, red) and about 7 small ones (green, yellow, blue, red)	Have
On/off switch	One switch to power the ball	1 round rocker switch, panel-mount	Have
Round shell	Holds and protects it all	Not yet	Need to buy

Part	Job	What we have	Status
Headbands x4	Show each player's color	Not yet	Need to buy
Power bank	Powers everything	1 white power bank	Have

Also on the bench, not needed for v1

Part	Possible use	Status
12-key number keypad	Number entry if we bring keypad puzzles back	Spare
Capacitive touch board (12 inputs, I2C)	Turn patches of the shell into touch pads instead of buttons. Easier to fit on a sphere	Spare
8x8 LED dot matrix with driver	Icons, a skull on explosion, simple animations	Spare
BBC micro:bit V2	Bench testing shake and drop detection now, before the accelerometer arrives. Later, the mic for the shout puzzle	Spare
Arduino-compatible board on a screw-terminal shield	Not needed. The ESP32 does it all	Spare
Breadboard, jumper wires, cardboard button panel	Bench prototyping before everything goes in the shell	Spare

Parts we've identified so far



Photo 3.1 · Rotary encoders (2). Spin freely and press like a button. One is the ball's knob, one is a spare.

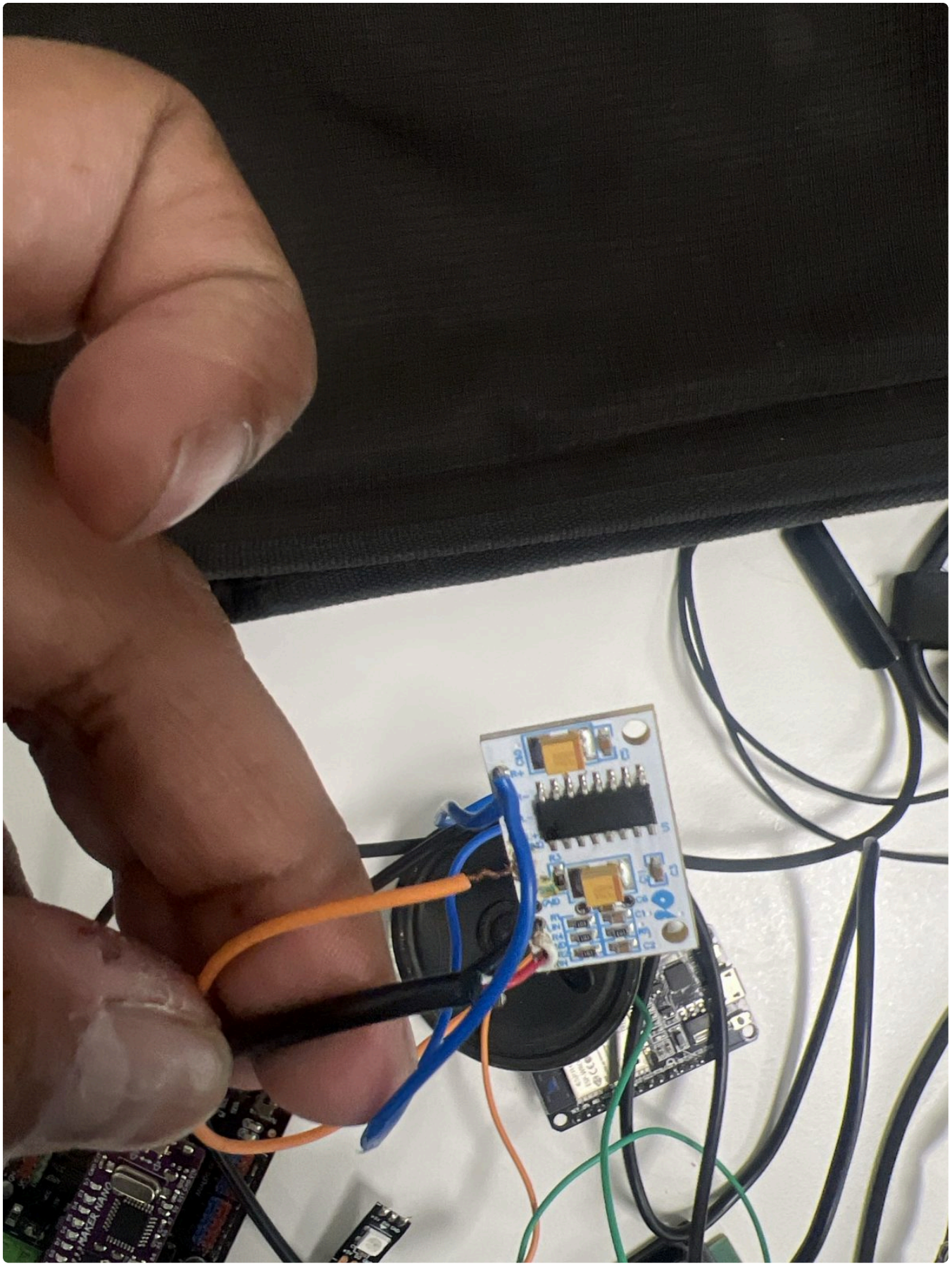


Photo 3.2 · PAM8403 amplifier, already wired to the speaker. Up to about 3W per channel on 5V.

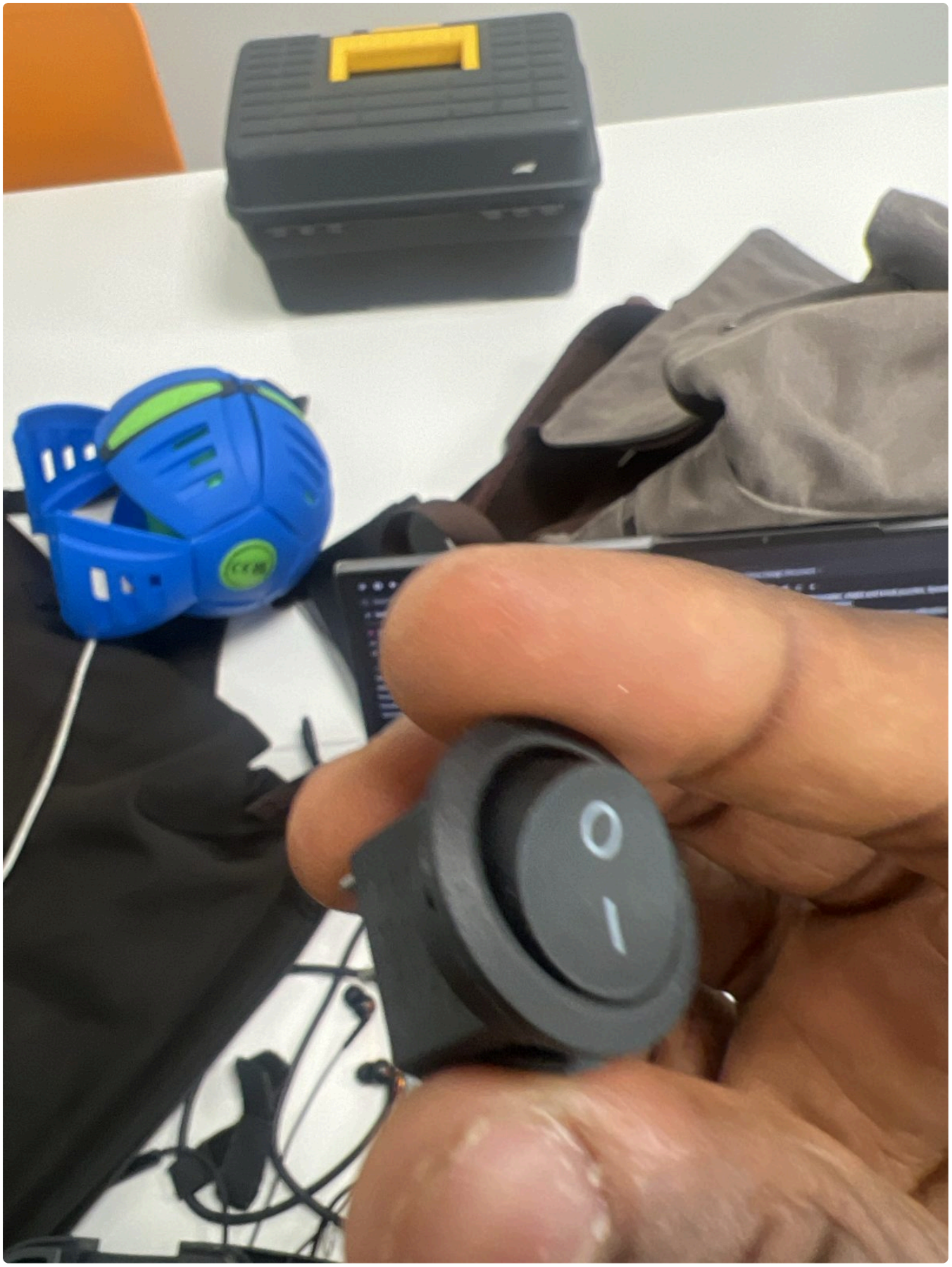


Photo 3.3 · Round rocker switch. The ball's single on/off switch.



Photo 3.4 · BBC micro:bit V2 (spare). Useful for bench-testing shake and drops, and later the shout puzzle.



Photo 3.5 · PIR motion sensor (spare). Senses people nearby, not falls. A possible attract mode later.

Proposed screen setup with what we have: the 16x2 LCD shows puzzles and menus, and one alphanumeric display shows the countdown. Both use I2C, so they share the same two pins with the accelerometer and touch board.

What this changes in the puzzles: neither screen shows colors. So maths and number sequence answers go in 4 positions on the LCD, left to right, matching the button order instead of button colors. Color sequences still play on the LED strip.

Headbands: no ESP32 needed

In v1 each headband is just a battery LED in a fixed color. The ball does all the signalling: when it tags a color, the screen floods that color, the LED ring matches it and the speaker shouts the color name.

- **What we gain:** zero extra boards, nothing to pair or charge beyond coin cells, setup takes seconds
- **What we lose:** headbands can't flash when tagged. The voice callout has to be loud enough to carry
- **Upgrade path:** if headbands that light up turn out to matter, each gets a small ESP32-C3 and the ball broadcasts to them directly. Still no base station

4. Setup and calibration

Setup is done entirely on the ball: one switch, then the knob to scroll and Enter to select on its screen. Calibration is split into a once-a-day hardware check and a short check per mode.

4.1 Setup

1. **Switch on the ball.** It boots straight to the main menu on its screen and shows battery level
2. **Pick a mode** with the knob, then Enter: Solo, Teams or Individual
3. **Teams and Individual:** pick the number of players and which headband colors are in play
4. **Settings load from last time.** Change any of them in the host menu (section 13)
5. **Run the mode check** (4.3 or 4.4)
6. **Set the ball on the ground and hold Enter** to arm it. A hold, not a tap, so it can't start by accident

Host menu: opened with a long press on Enter from the main menu, so players can't change settings mid-game. Everything set there is saved on the ball and survives a power-off.

4.2 Hardware calibration (all modes, once per day)

Run at the start of the day, after moving venues, or after a battery swap. The ball walks the host through each step on screen and saves the results.

Step	What the host does	What the ball learns or checks
Buttons	Press each button once	Every button registers. Screen shows its color
Knob check	Turn a few clicks each way, then press the knob	Turning direction is right, every click counts and the push registers. No range to set, since an encoder spins freely
Shake threshold	Hold the ball still, then pass it normally, then shake it hard	Sets the shake trigger above normal handling and below a real shake
Pickup	Set the ball down, then lift it	Ground and pickup detection, used to arm and start rounds
Screen	Watch a full-color sweep	Dead pixels or a cracked window
LED ring	Watch each color in turn	All LEDs and colors work
Vibration	Feel the short, medium and long buzz	Motor works and is strong enough to feel
Volume	Play a test callout and explosion, turn the knob to set level	Loud enough for the room. Saved as the default
Battery	Read the level on screen	Swap the power bank if it's low

4.3 Solo calibration (before each session)

Takes about a minute. Tunes the challenge to the crowd.

1. **Choose a difficulty preset:** Easy, Normal or Hard. Each sets countdown length, puzzles to defuse and bonus time per solve. The notes' starting point is 5 minutes and 10 puzzles
2. **Choose the puzzle mix:** turn any of the five puzzles on or off
3. **Practice puzzle:** the ball runs one puzzle of each active type with the timer paused, so the host can demo them
4. **Adjust after real rounds:** if most groups win easily, lower bonus time or raise puzzles to defuse. If most lose early, do the opposite

4.4 Teams calibration (before each match)

Takes one to two minutes. Makes sure every player knows their color and can hear the ball.

1. **Colors in play:** on the screen, turn each headband color on or off and assign it to Team A or Team B with the knob
2. **Hand out and switch on headbands**
3. **Callout check:** the ball calls every active color in turn, on screen and by voice. Each player raises a hand when they hear theirs. If someone can't hear it, turn up the volume or move closer together
4. **Pass risk:** choose 1 or 2 dark buttons out of 4
5. **Timer:** starting length, how much it shrinks per reset, and the minimum
6. **Points target:** how many points win the set

4.5 Individual calibration (before each game)

Same as Teams, minus the team split.

1. **Colors in play:** turn each headband color on or off. Each color is one player
2. **Hand out and switch on headbands**
3. **Callout check:** the ball calls every active color. Each player raises a hand when they hear theirs
4. **Pass risk:** choose 1 or 2 dark buttons out of 4
5. **Timer:** starting length, how much it shrinks per turn, and the minimum

4.6 Imposter calibration

Not designed yet. It will reuse 4.2 and 4.4, plus a step to secretly assign the imposter.

5. Game modes overview

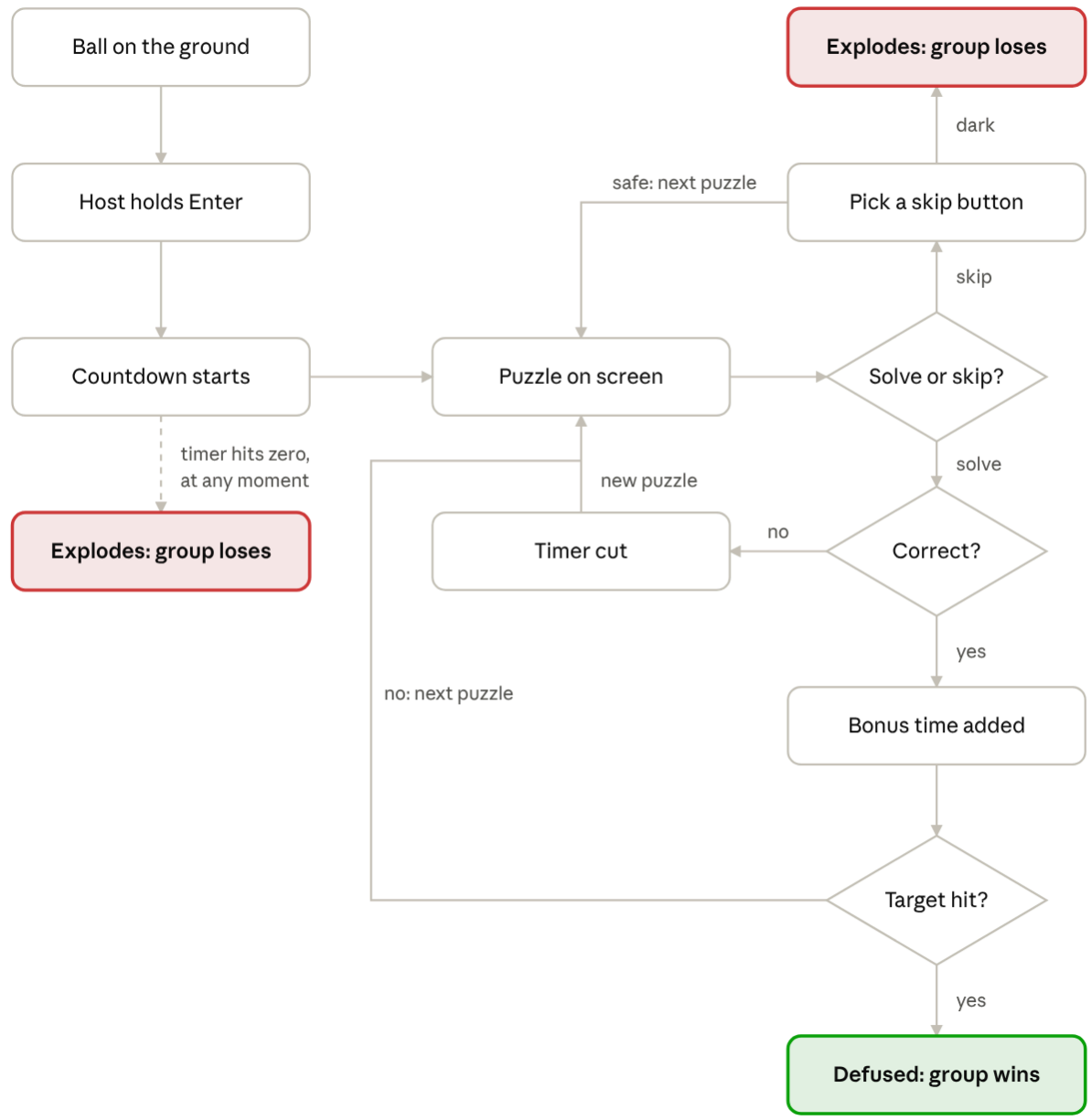
There are four modes. Solo is built first because it needs no headbands and tests the core fun on its own.

	Solo (Rapid Fire)	Teams	Individual	Imposter
Status	v1, build first	v1, build second	v1, after Teams	Parked
Players	Everyone around one ball	2 to 8, split into two teams	Each player for themselves	Group with one hidden imposter

	Solo (Rapid Fire)	Teams	Individual	Imposter
Headbands	None	Yes	Yes	Yes
Pass or skip	Skip a puzzle, with dark button risk	Pass to the other team, any time	Pass to another player, any time	Ball passed round for voting
Scoring	Solve the target number to defuse	+1 per solve, -1 per explosion (never below 0)	Explode and you're out	Vote out the imposter
Win	Group defuses the bomb	First team to the points target wins the set	Last player standing	Civilians find the imposter
Lose	Timer runs out, or a dark button	Opponent reaches the target first	Knocked out by an explosion	Timer runs out first
Best for	Demos, short lines, first-timers	Competitive groups, spectators	Longer sessions, friend groups	Groups who stay a while

6. Solo mode (Rapid Fire)

Everyone plays together on one ball, with no headbands. Stuck on a puzzle? You can gamble on a skip. Solve the target number of puzzles before the timer runs out. The notes' starting point is 10 puzzles in 5 minutes, both tunable.



Solo mode flow · solve loop, two endings

The group loops through puzzles until they hit the target or the bomb explodes. A wrong answer only costs time. A bad skip costs the game.

Rules

1. The ball starts on the ground. The host holds Enter and the countdown begins
2. A puzzle appears on the screen, picked from the active puzzle mix
3. Whoever is holding the ball solves it and presses Enter. Anyone can grab the ball and help
4. **Correct:** bonus time is added and the counter goes up

5. **Wrong:** the timer is cut, then a new puzzle appears
6. **Skip:** instead of solving, pick one of the pass buttons. A safe button skips to the next puzzle with no time lost. A dark button explodes the bomb and the group loses
7. Hit the target number of solves: the bomb is defused and the group wins
8. Timer hits zero at any point: the bomb explodes and the group loses

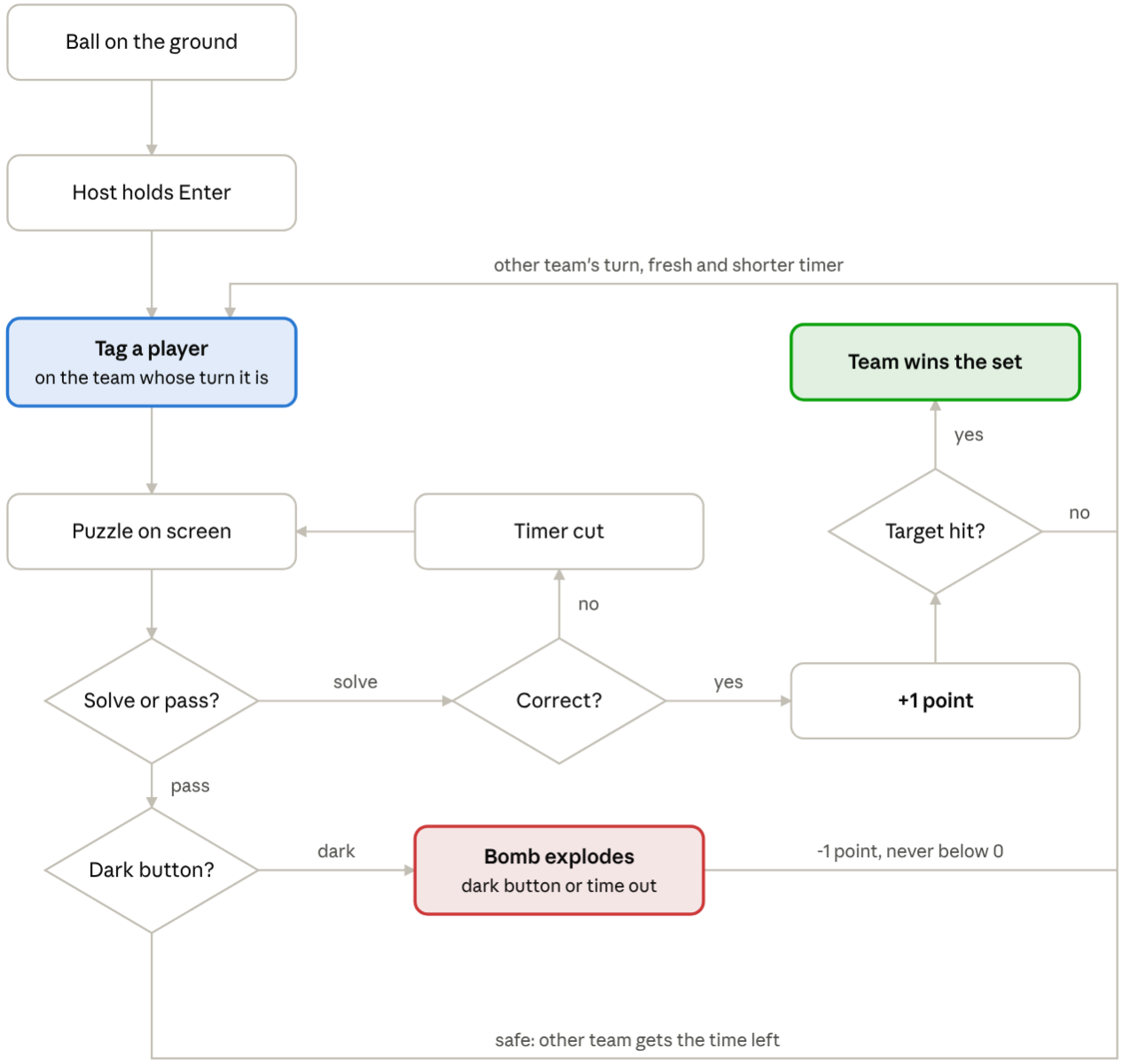
Why it's built first: it needs only the ball, it runs with any group size, and it proves whether the puzzles are fun before any headband exists.

7. Teams mode

Two teams take turns with the bomb and score points by solving puzzles. Passing dumps the bomb on the other team but risks blowing it up in your hands. First team to the points target wins the set.

Teams: solve to score, pass to dump the bomb and risk a dark button

Teams take turns. First team to the points target wins the set.



Teams mode flow · tag, then solve or gamble on a pass

Every path ends with the other team's turn. A new round gets a fresh timer that's a little shorter than the last. A safe pass doesn't: the other team inherits whatever time is left.

Rules

1. 2 to 8 players split into two teams, each wearing a headband in a fixed color
2. The ball starts on the ground. The host holds Enter to start the set
3. Teams take turns. Each round, the screen tags a player on the team whose turn it is
4. A fresh timer starts, slightly shorter each round as the set goes on
5. The tagged player can **solve** or **pass** at any time. Passing without trying first is allowed
6. **Correct solve:** their team scores +1. The next round goes to the other team

7. **Wrong answer:** the timer is cut. Try again, or pass
8. **Pass:** pick one of the 4 physical buttons. Safe means the other team takes the bomb with the time left. Dark means it explodes on the passer
9. **Explosion** (timer runs out, or a dark button): that team loses 1 point, never below 0. The next round goes to the other team
10. **First team to the points target wins the set.** One set is one match

Strategy this creates

- Solving is the only way to score, so teams want to try
- Passing early hands the other team a healthy timer and a good shot at the point. Passing late hands them a nearly dead bomb
- A bomb passed back and forth gets more dangerous, because every wrong answer cuts the timer and the cut carries over
- Teammates shout answers and opponents shout wrong ones

Fairness: puzzle difficulty is balanced across both teams, with some randomness. Exact puzzle sets get worked out in playtesting.

7b. Individual mode

Every player for themselves. Explode and you're out, and the last player standing wins. There are no sets in this mode: it's one elimination game, so it runs longer than Teams.

Rules *(proposed, to confirm in playtesting)*

1. Each player wears their own headband color
2. The ball tags a random player, who must solve or pass
3. **Correct solve:** the bomb is tagged to another player with a fresh, slightly shorter timer
4. **Wrong answer:** the timer is cut. Try again, or pass
5. **Pass:** pick a button. Safe means the bomb goes to another player with the time left. Dark means it explodes on the passer
6. **Explosion** (timer runs out, or a dark button): that player is out and their color leaves play
7. **Last player standing wins**

Still open: after a correct solve, does the bomb go to a random player or does the solver choose? And how many headbands does it need to be fun?

8. Core mechanics

Four mechanics run the whole game: tagging, solving, the pass gamble and the explosion. Rules marked *proposed* fill gaps in the notes and need confirming.

8.1 Tagging

The game calls a color and the player wearing it must grab the ball.

- The callout fires on every feedback channel at once: the ball's screen floods with the color, the speaker shouts it and the LED ring turns that color. Headbands don't change, so the voice and the screen are what the tagged player catches
- The tagged player picks up the ball and the puzzle appears
- Tagging exists in Teams, Individual and Imposter. Solo has no headbands, so everyone works the same ball

8.2 Solving

The holder solves the puzzle shown on the screen using the ball's inputs, then confirms.

- **Solo:** each solve adds bonus time. Solve the target number to defuse the bomb
- **Teams:** each solve scores +1 for the team. The next round goes to the other team with a fresh, shorter timer
- **Individual:** a solve sends the bomb to another player with a fresh, shorter timer
- **Wrong answer (all modes):** the timer is cut. Try again, or pass or skip

8.3 The pass gamble (all modes)

Passing is the escape route, and it's always a gamble. It works the same in every mode; only where the bomb goes changes.

- The holder can pass at any time, with or without trying the puzzle first
- They pick one of the **4 physical buttons**. One or two are secretly dark (a setting)
- **Safe button:** the bomb moves on with the timer as-is, no reset
- **Dark button:** the bomb explodes on the passer

Mode	Called	Safe button	Dark button
Solo	Skip	Next puzzle, no time lost	Bomb explodes, group loses
Teams	Pass	Other team takes the bomb with the time left	Passer's team loses 1 point

Mode	Called	Safe button	Dark button
Individual	Pass	Another player takes the bomb with the time left	Passer is out

Why physical buttons: a real button press gives better feedback and more pressure than picking on a screen. The 3 by 3 on-screen grid idea is dropped.

8.4 Explosion

The bomb explodes when the timer hits zero, or when a dark button is picked during a pass.

- Full-channel punishment: loud explosion audio, LED strip strobes, strong vibration, screen shows the explosion and a mockery line
- **Solo:** the group loses
- **Teams:** the team holding it loses 1 point, never below 0. The next round goes to the other team
- **Individual:** the holder is out. Last player standing wins

9. Puzzle catalog

Five puzzles ship in v1, all solved on the ball and confirmed with an Enter button. They fall into two groups: brain puzzles (defusal) and physical puzzles (manipulation).

Ball inputs in v1: 4 colored buttons, 1 Enter button, 1 knob (rotary encoder that also presses like a button), 1 accelerometer.

Puzzle	Group	What the screen shows	What the player does	Ball input	Status
Maths	Brain	An addition or subtraction sum, plus 4 answers in a row matching the button order	Press the button in the right answer's position, then Enter	Colored buttons + Enter	v1
Number sequence	Brain	A sequence with a gap, plus 4 options in a row matching the button order	Press the button in the missing number's position, then Enter	Colored buttons + Enter	v1
Color sequence	Brain	"Watch the lights" while the LED strip flashes a color sequence once	Repeat it on the colored buttons, then Enter	Colored buttons + Enter	v1

Puzzle	Group	What the screen shows	What the player does	Ball input	Status
Knob to position	Physical	A dial with a target marker	Turn the knob until the needle on screen lines up with the target, then press Enter. The needle starts in a random spot each time	Knob + Enter	v1
Shake the ball	Physical	A shake meter that fills up	Shake hard until the meter is full	Accelerometer	v1
Shout at it	Physical	A volume meter	Shout until the meter fills	Microphone	Cut for v1

Changes from the original notes:

- **No keypad.** Maths and number sequence were planned on a keypad. Showing four color-coded answers on screen lets the same 4 buttons handle three puzzles, and a keypad on a sphere is hard to build
- **Shout at it is cut.** Festival floors are loud and everyone is already shouting, so a mic can't reliably tell a player's shout from the crowd. It also needed a micro:bit as a second platform. Revisit with a mic module on the ESP32 later
- **Shake needs tuning.** The threshold must be high enough that passing the ball doesn't count as a shake

Difficulty levers (per puzzle): bigger numbers, longer color sequences, tighter knob tolerance, a fuller shake meter. Difficulty can rise as the round goes on.

Screen on the ball: every puzzle has to read at arm's length on a small screen. The 16x2 LCD fits one short question and four answers in a row, laid out in the same order as the buttons. The countdown lives on its own alphanumeric display so it never competes for space.

10. Feedback system

Every game event is shown on at least two channels, so players catch it even in a loud room. All four channels live on the ball.

Channels

- **LED screen (visual):** mounted on the ball behind a protective window
- **Speaker:** inside the ball, through an audio module
- **LED ring:** lights the shell from inside
- **Haptics:** vibration motor inside the ball

Which channel fires on which event (from the scope notes)

Event	Screen	Audio	LEDs	Haptics	Example
Mode selection	Yes				Menu highlights the chosen mode
Instruction	Yes	Yes			"Grab the ball and solve the sum!"
Tagging (color callout)	Yes	Yes	Yes		Screen floods blue, voice shouts "Blue!", LED ring turns blue
Countdown	Yes	Yes	Yes	Yes	Ticking speeds up, LED ring drains, pulses get faster
Warning (last seconds)	Yes	Yes	Yes	Yes	Red flashing, alarm beeps, strong buzzing
Input feedback	Yes	Yes		Yes	Short click and buzz on each press, correct or wrong sound
Intimidation / mockery	Yes	Yes			"Your teammate is not going to save you"
Punishment / explosion	Yes	Yes	Yes	Yes	Boom, white strobe, long hard buzz, explosion on screen
Interface visualization	Yes				Live puzzle, timer and team status

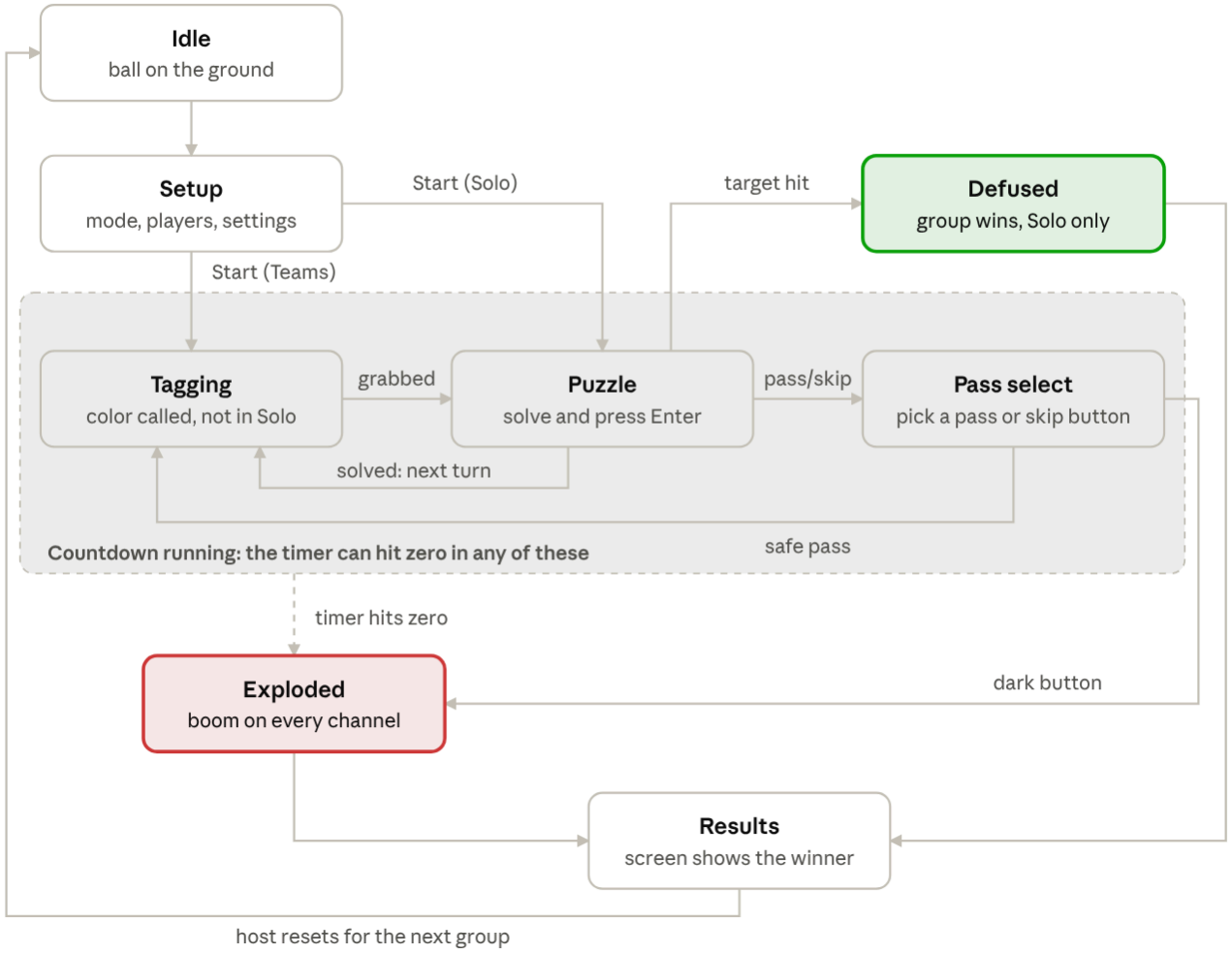
Notes

- Headbands are static in v1, so they're not a feedback channel. They only show who is which color
- The screen is small and faces one way. Spectators follow the game by sound and the LED ring, so those two carry the drama
- Mockery lines are pure content. They're cheap to write and record, and they're a big part of the personality
- A speaker inside a padded ball loses volume. Test it in a noisy room early

11. Game logic: state machine

All modes share one state machine running on the ball's ESP32. Solo skips Tagging, and a safe skip goes straight back to Puzzle. Teams and Individual never reach Defused.

Every round moves through eight states, all run inside the ball



game state machine · 8 states, both modes

The three states inside the dashed zone are where the countdown runs, so a timeout can fire from any of them.

State	Screen	Speaker	LED ring and vibration
Idle	Main menu, battery level	Quiet	Soft glow
Setup	Mode, players, settings, calibration steps	Menu clicks, test sounds	Test patterns during calibration
Tagging	Floods with the tagged color	Shouts the color	Ring turns that color
Puzzle	Puzzle, timer, answer blocks	Ticking that speeds up, right and wrong sounds	Ring drains with the timer, buzz on each press

State	Screen	Speaker	LED ring and vibration
Pass select	The pass buttons in their colors	Tense loop	Ring flickers
Exploded	Explosion and a mockery line	Explosion, then mockery	White strobe, long hard buzz
Defused	Win screen	Win sound	Green, short happy buzz
Results	Who won, "hold Enter to reset"	Quiet	Soft glow

12. Firmware structure and wiring

With one ESP32 there's no network protocol to design. The firmware is a set of modules inside one program, and every part is wired straight to the board.

Firmware modules

Module	Job
Input	Reads buttons, knob and accelerometer. Turns raw readings into events: button pressed, knob moved, shake, picked up, set down
Game	The state machine (section 11). Owns the timer, picks puzzles, checks answers
Puzzles	One small file per puzzle type, all with the same shape: start, handle input, report solved or failed
Display	Draws menus, puzzles, timer and callouts on the screen
Audio	Tells the audio module which sound to play and at what volume
Lights	LED ring patterns: glow, drain, flash, strobe
Haptics	Vibration patterns: tick, buzz, long
Settings	Saves settings and calibration to the ESP32's flash so they survive power-off

The game module never touches hardware directly. It only sends requests like "play explosion" or "show puzzle", which keeps puzzles easy to add and tweak.

Wiring: what connects to the board

Part	Connection type
4 colored buttons + Enter	5 digital input pins
Knob	3 digital pins: 2 for turning, 1 for its push button
Accelerometer	I2C (2 pins, shared bus)
LED screen	LCD and alphanumeric display both on the I2C bus
LED ring	1 data pin
Vibration motor	1 pin, through a transistor
Audio module	Serial (2 pins) for an MP3 module, or I2S (3 pins) for an amplifier

That's roughly 12 to 16 pins in total, which a standard ESP32 dev board has room for. The LCD, alphanumeric display, accelerometer and touch board can all share one I2C bus, which keeps the pin count low.

If smart headbands are added later: the ball broadcasts tag and explosion messages over ESP-NOW straight to each headband's ESP32-C3. The ball stays the only brain.

13. Tunable parameters

Every timing and difficulty value is a setting the host changes in the ball's host menu, so the game can be tuned during playtests without reflashing. The notes' starting point is 10 puzzles in 5 minutes for Solo.

Setting	Mode	Starting value	Notes
Countdown length	Solo	5 min	From the notes. Tune in playtest
Puzzles to defuse	Solo	10	From the notes
Bonus time per solve	Solo	Set in playtest	Higher makes Solo easier
Starting timer	Teams, Individual	Set in playtest	Fresh at the start of each round or turn
Timer shrink per round	Teams, Individual	Set in playtest	Keeps late rounds tense

Setting	Mode	Starting value	Notes
Minimum timer	Teams, Individual	Set in playtest	Floor so it never becomes impossible
Wrong-answer time cut	All	Set in playtest	Seconds taken off. Carries over on a safe pass
Pass or skip risk	All	1 dark button of 4	Raise to 2 of 4 if teams pass too often
Points target	Teams	Set in playtest	First team to reach it wins the set
Explosion penalty	Teams	-1 point	Scores never go below 0
Puzzle mix	All	All five on	Turn puzzles on or off
Puzzle difficulty	All	Easy	Balanced across teams, with some randomness
Shake threshold	All	Set in calibration	Too low and passing counts as a shake
Knob tolerance	All	Set in playtest	How close counts as on target
Volume	All	Set in calibration	Check in a loud room

14. Imposter mode (parked)

Imposter mode is out of v1 scope. This section only records what's in the notes so the idea isn't lost.

From the notes:

- Uses headband colors
- The ball is passed around the group
- Everyone votes on the color of the imposter
- If the voted player is **not** the imposter: the timer halves and the voted player leaves
- If the voted player **is** the imposter: the civilians win

To design later:

- How the imposter is told secretly, and what their goal is
- What the imposter can do to sabotage (for example, wrong answers that look like honest mistakes)
- How votes are cast on the ball

- What the imposter wins on

Related idea, also parked: a rapid-fire skip in Solo that lets the group skip a puzzle at a risk. The notes mention it but don't define the risk.

15. One-week build plan

Solo is finished and playtested by day 5, so a complete game exists even if Teams slips. Everything is built on the one ESP32 in the ball.

Day	Build	Done when
1	ESP32 on the bench with buttons, knob and screen	Pressing a button or turning the knob shows on the screen
2	Solo mode state machine. Maths, number and color sequence puzzles	A full Solo round can be won and lost on the bench
3	Accelerometer, shake and knob puzzles. Speaker, LED ring, vibration	All five puzzles and all four feedback channels work
4	Host menu, saved settings, hardware calibration (4.2) and Solo check (4.3)	The ball can be set up and calibrated with no computer attached
5	Shell with protected screen window. Playtest Solo with real people	Strangers can play without help after one explanation
6	Teams mode: tagging, pass gamble, Teams check (4.4). Static headbands	A full Teams match can be played
7	Individual mode if there's time (it reuses Teams logic). Fixes, tuning, volume and battery check	Runs for hours on one power bank swap

Risks to the plan

- **Parts arriving late.** If parts aren't in hand by day 2, the week is gone. Source locally or next-day
- **First time with ESP32 or soldering:** add 1 to 2 days. Wiring and enclosures eat time, not code
- **The enclosure.** A round shell with button holes is fiddly. Have a backup shell ready

16. Open questions, assumptions and backlog

Decided

- Teams: solve scores +1, explosion costs -1 (never below 0), rounds alternate, first to the points target wins the set
- Teams: passing is allowed at any time, and a safe pass keeps the timer as-is
- Pass buttons: the 4 physical buttons, not an on-screen grid
- Solo: a wrong answer cuts time; a skip on a dark button explodes the bomb
- Individual is its own mode, with elimination and no sets

Still open

- [ ] Individual: after a correct solve, does the bomb go to a random player or does the solver choose?
- [ ] Individual: how many headbands does it need to be fun?

Parked until after prototyping

- [ ] Game name for the event listing, styled so it reads clearly as a toy
- [ ] Screen setup: LCD for puzzles and an alphanumeric display for the timer, or something else? Also confirm the part counts in section 3, which were read from photos
- [ ] Where on the ball does the screen sit, and how is it protected? Padding, recessed window or a clear cover
- [ ] Is the voice callout loud enough to replace headbands that light up? Test in a noisy room

Assumptions this document makes

- One ESP32, inside the ball, runs everything. No laptop, no base station
- The LED screen sits on the ball, protected against drops
- Headbands are static battery LEDs in fixed colors
- The ball is passed by hand, never thrown
- Who is holding the ball is judged by eye. The ball can't detect its holder
- Sounds play from an audio module, not generated by the ESP32

Backlog (after v1)

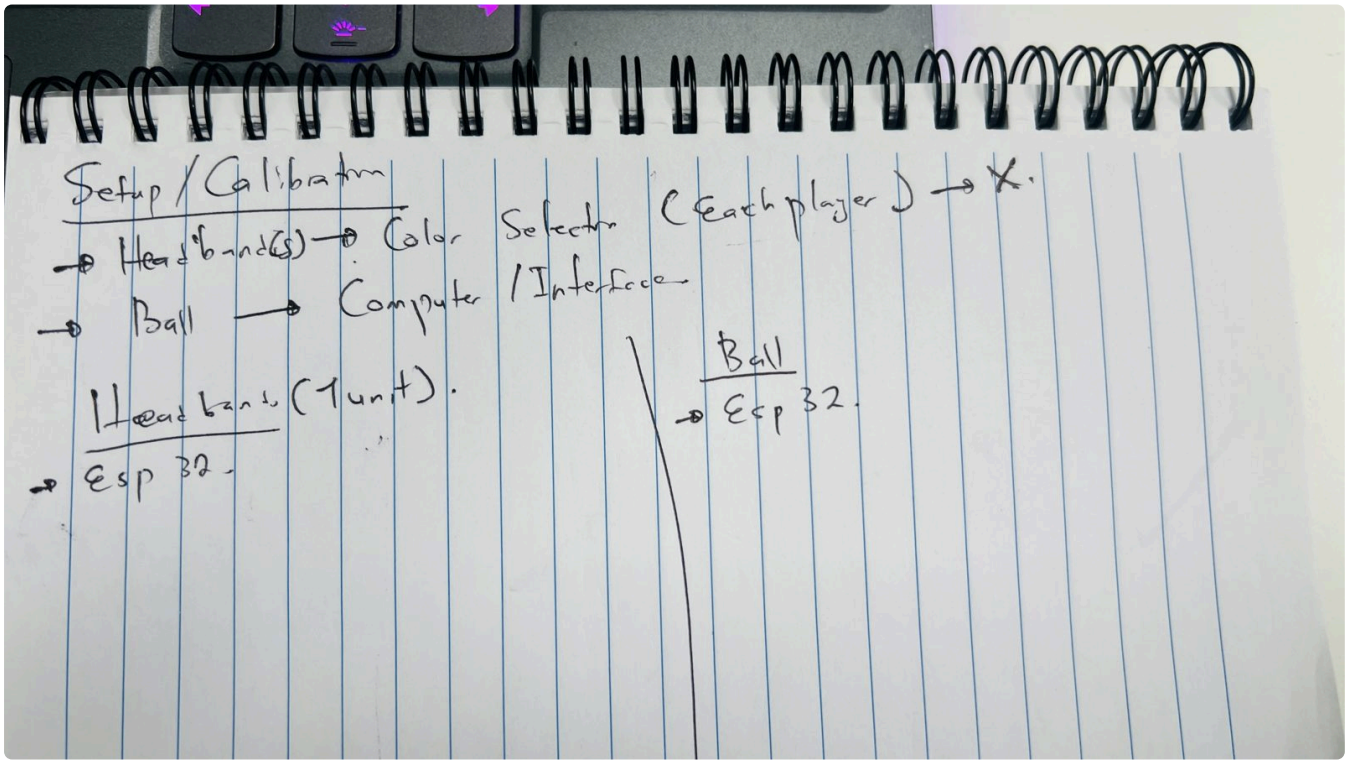
- Imposter mode
- Shout at it puzzle with a mic module
- Smart headbands that light up when tagged (ESP32-C3 each, ESP-NOW from the ball)
- Per-player color selection
- Up to 8 headbands
- Throwable, impact-proof ball
- Custom batteries and charging
- Color-blind support: callout says the color name, plus a pattern on each headband

- A second, larger screen for spectators

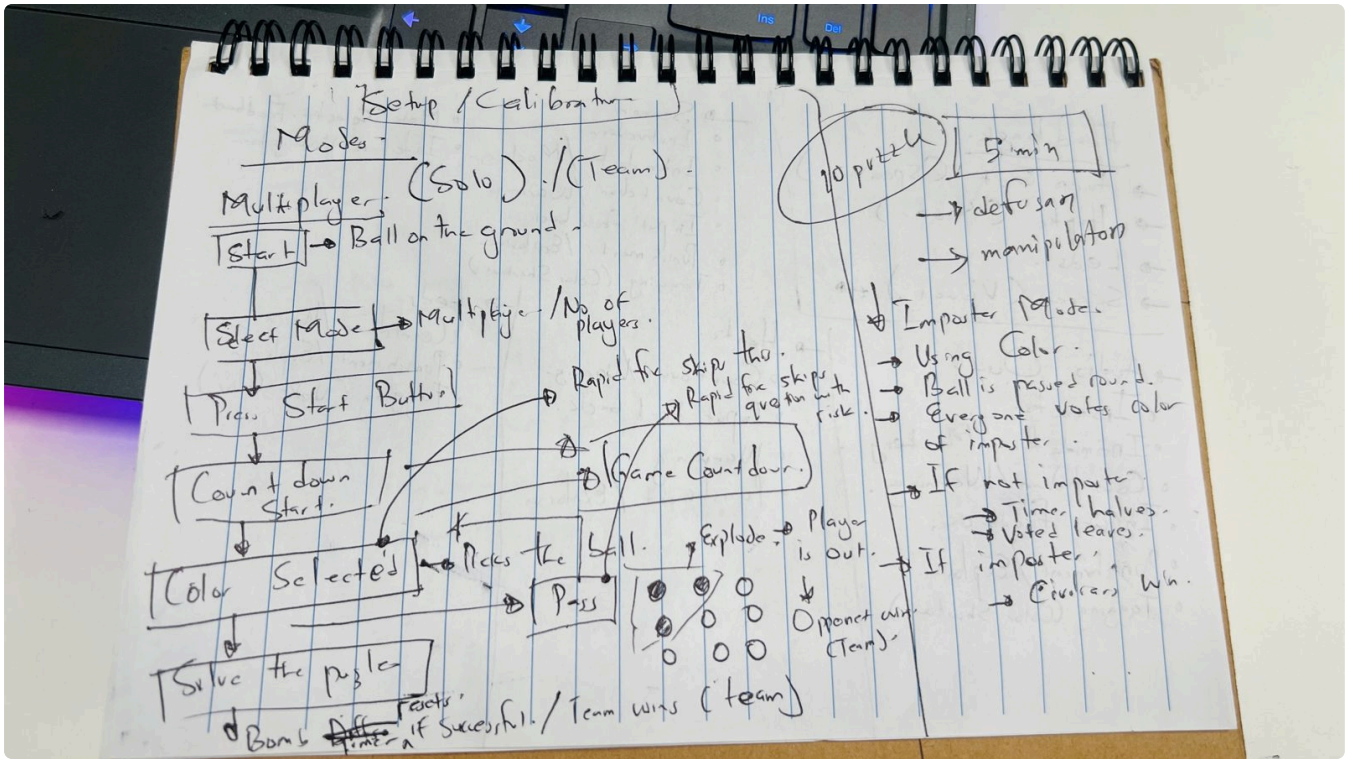
17. Sketches and prototype photos

The notebook pages this document was built from, and the bench prototype so far.

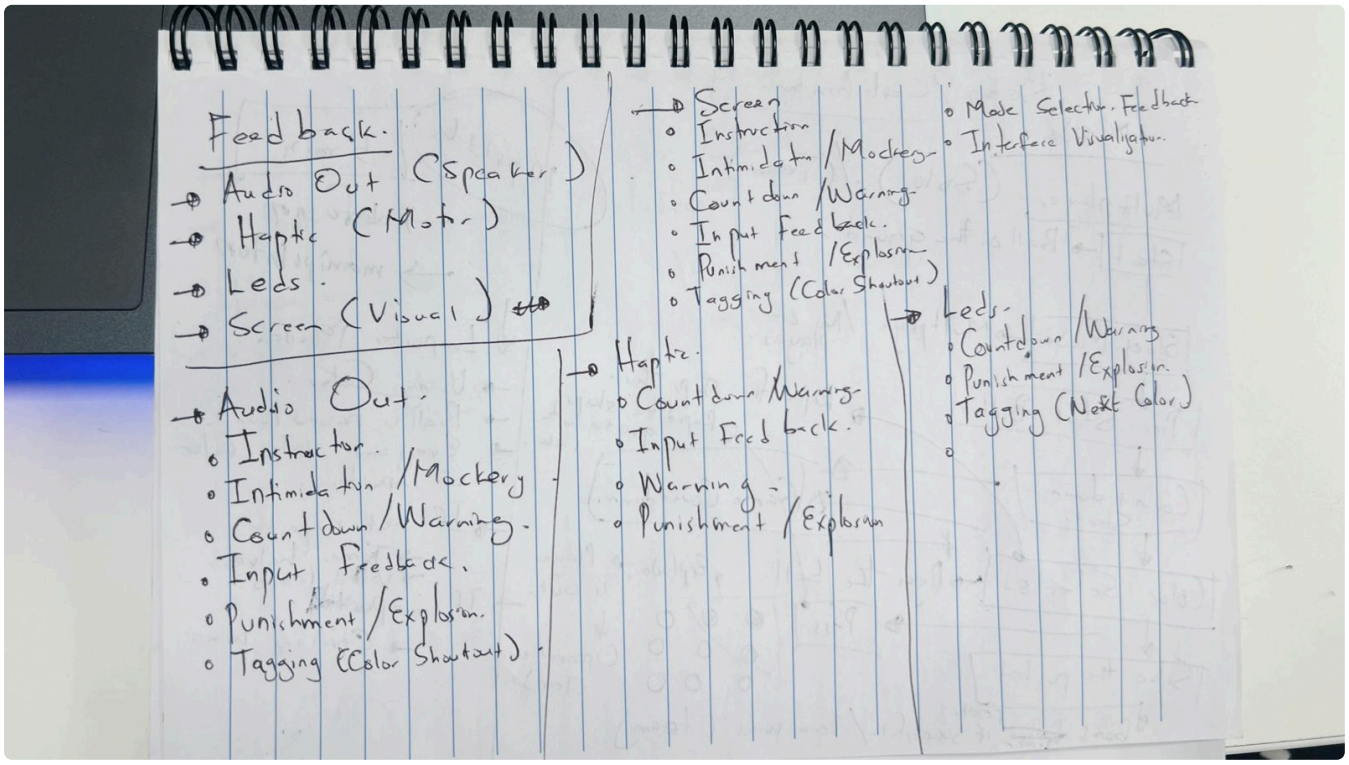
17.1 Design sketches



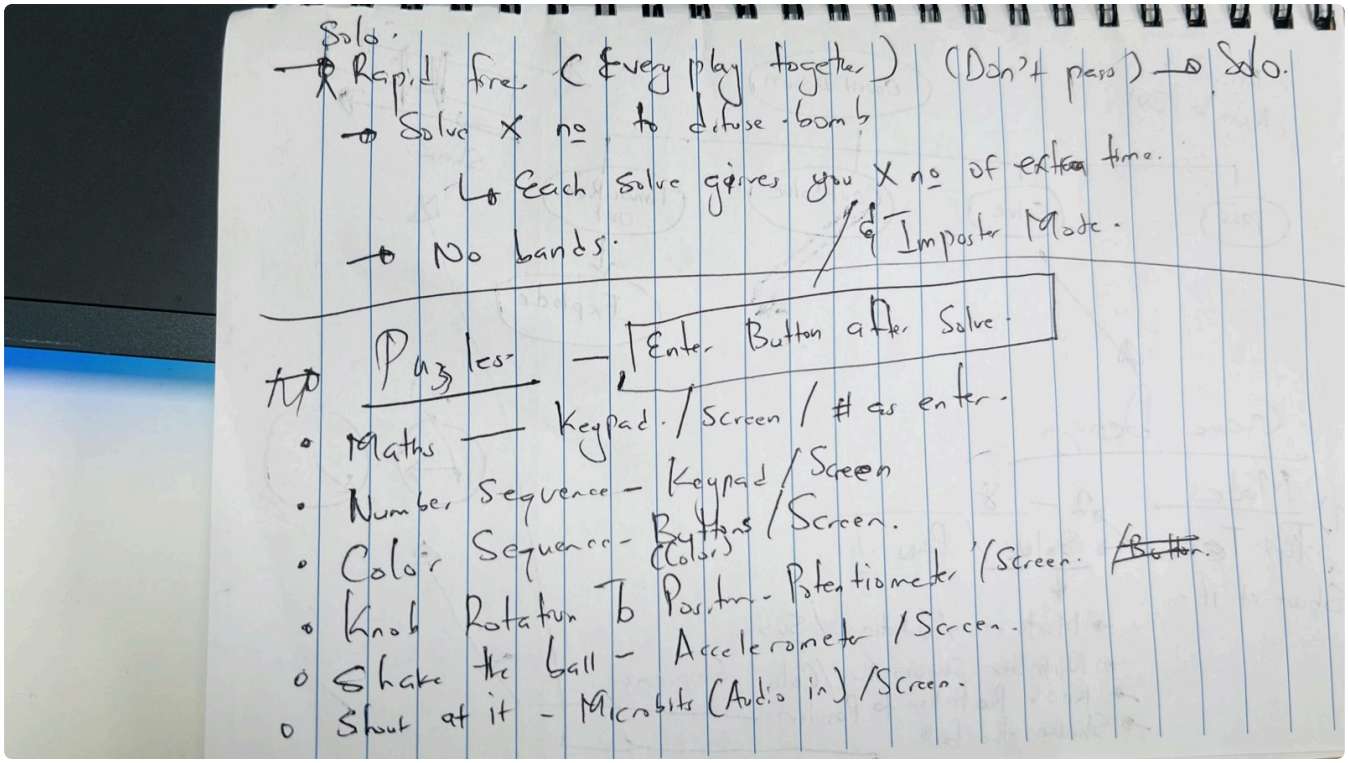
Sketch 1 · Setup and calibration. Headbands and ball each with an ESP32 (since cut to one ESP32 in the ball).



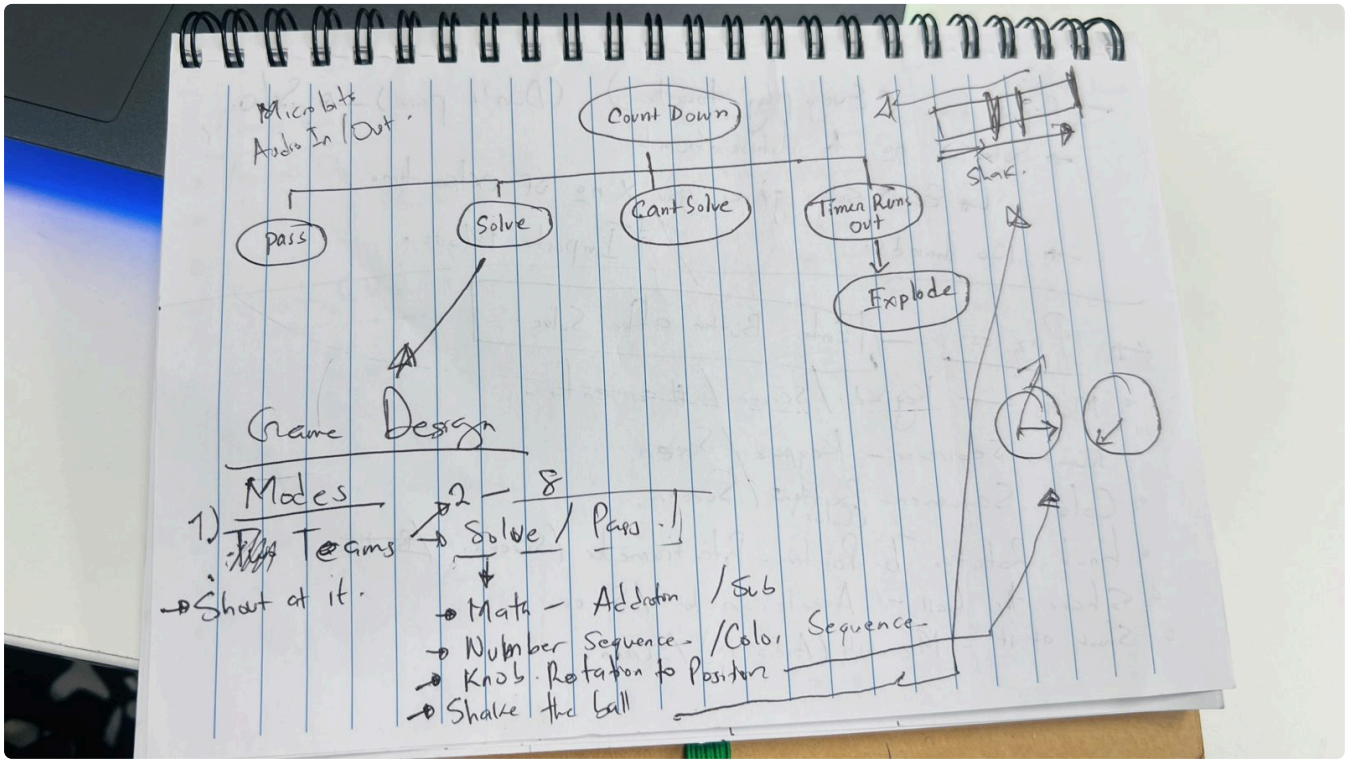
Sketch 2 · Game flow: modes, countdown, color selection, solving, the pass grid with dark buttons, and Imposter notes.



Sketch 3 · Feedback channels (audio, haptics, LEDs, screen) mapped to game events.



Sketch 4 · Solo mode and the puzzle list with the input each one needs.



Sketch 5 · Game states (countdown, solve, can't solve, pass, explode) and the Teams outline.

17.2 Bench prototype

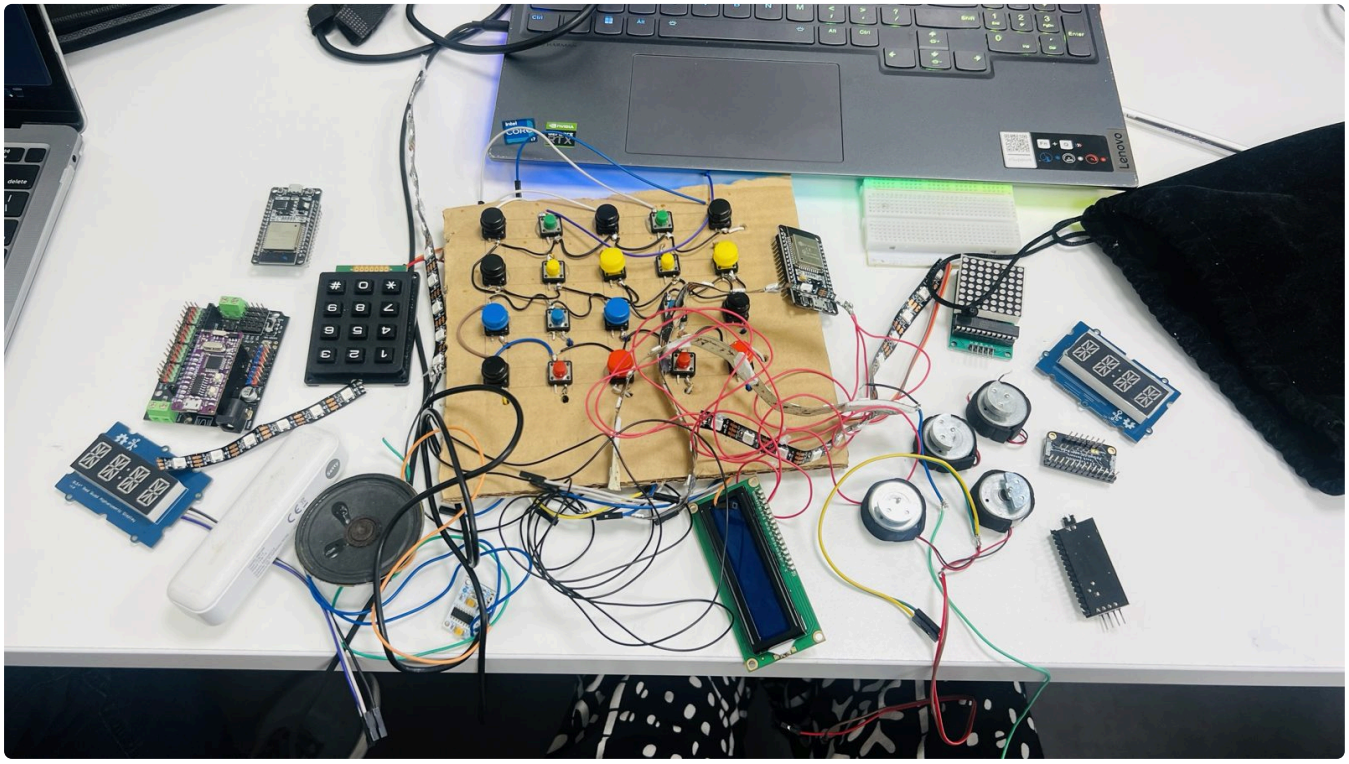


Photo 17.1 · Bench prototype: cardboard button panel wired to an ESP32, with the LCD, alphanumeric displays, LED strip, motors and speaker.

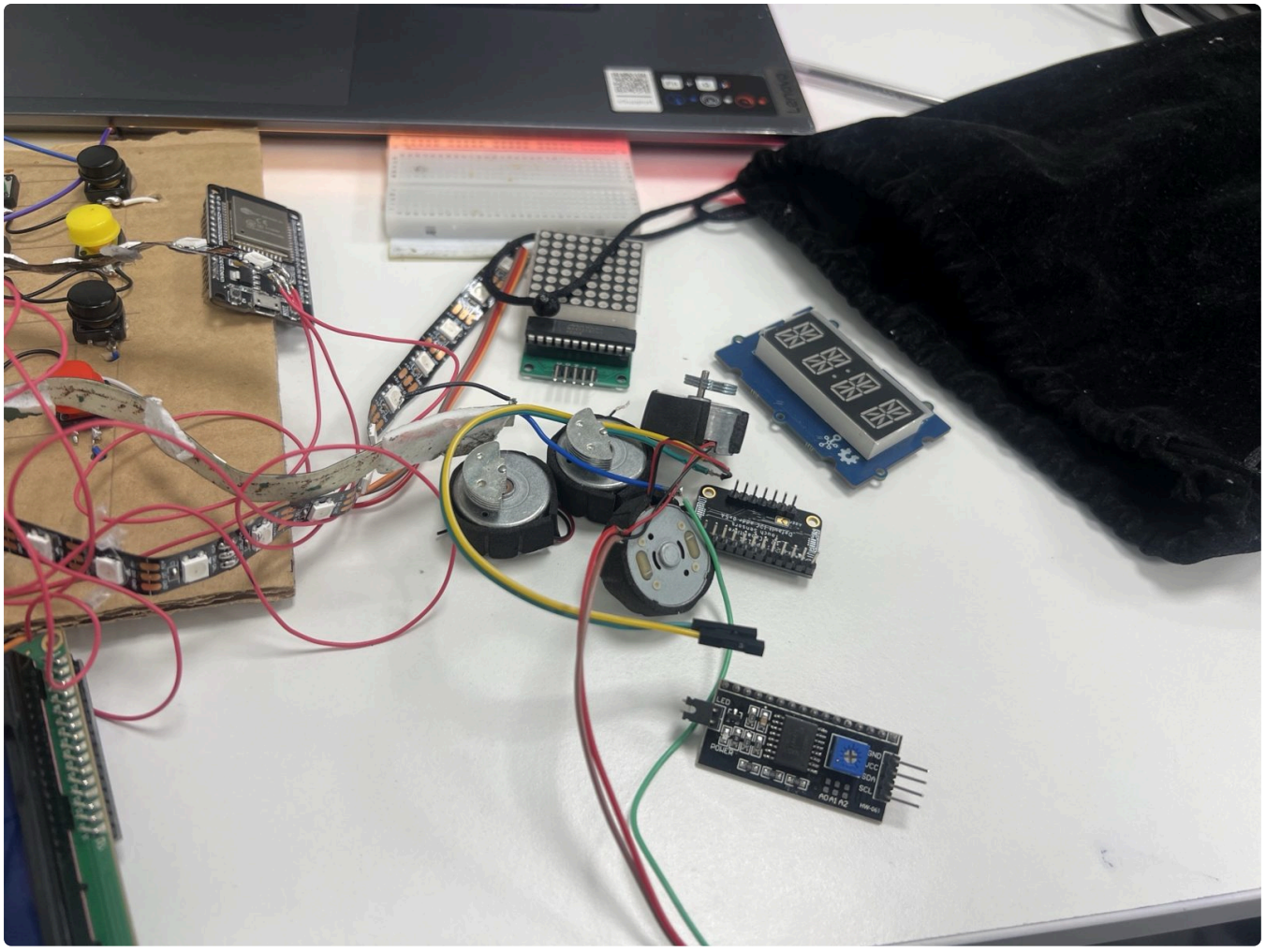


Photo 17.2 · Close-up: ESP32, LED strip, vibration motors, alphanumeric display and the LCD's I2C board.